THEOREM 3.12 OF CAMERON, GOETHALS, SEIDEL AND SHULT: STRUCTURED PROOFS

ABSTRACT. We characterize the sets of lines at angles 90°, 60° and 45° which lie in a finite reduced crystallographic Root System. A compatible colouring gives an explicit construction in an integer span. Conversely, the parity of the integer logarithm of squared root length gives a compatible colouring. We include the empty set of lines and sets of lines which do not span V.

INPUTS FROM THE BLUEPRINT

We will use the following definitions and identities from the blueprint. They are repeated only to make this document standalone. Throughout, $q(\mathbf{x})$ means $(\mathbf{x},\mathbf{x})$, and $\mathbb{R}\mathbf{x}$ means $\mathrm{span}_{\mathbb{R}}\{\mathbf{x}\}$.

Definition 3.9. ASSUME: V is a finite-dimensional real inner product space.

A *Root System* in V is a finite set $\Phi \subseteq V \setminus \{0\}$ with $\mathrm{span}_{\mathbb{R}}\,\Phi = V$ such that, for all $\mathbf{r},\mathbf{s} \in \Phi$ and all $t \in \mathbb{R}$,

$$\frac{2(\mathbf{r},\mathbf{s})}{q(\mathbf{r})} \in \mathbb{Z}, \qquad \tau_{\mathbf{r}}\mathbf{s} := \mathbf{s} - \frac{2(\mathbf{r},\mathbf{s})}{q(\mathbf{r})}\mathbf{r} \in \Phi, \qquad t\mathbf{r} \in \Phi \implies t \in \{1,-1\}.$$

We write IsRootSet(Φ) for this predicate. The empty set is permitted when $V = 0$. This is the finite, reduced, crystallographic convention of the corrected Definition 3.9.

Definition (line systems of Section 2). ASSUME: V is a finite-dimensional real inner product space.

A line system R is a finite set $R.\Phi \subseteq V$ satisfying

$$R.\Phi = -R.\Phi, \qquad q(\mathbf{r}) = 2 \quad (\mathbf{r} \in R.\Phi), \qquad (\mathbf{r},\mathbf{s}) \in \{-2,-1,0,1,2\} \quad (\mathbf{r},\mathbf{s} \in R.\Phi).$$

It need not span V.

Elementary identities. We will use bilinearity over $\mathbb{R}$, symmetry, $q(\mathbf{x}) \geq 0$, and $q(\mathbf{x}) = 0 \iff \mathbf{x} = 0$. These are the facts of `blueprint/sec2-lines.tex`, repeated here for the standalone build. Their declarations in `Mathlib/Analysis/InnerProductSpace/Basic.lean` are `inner_add_left`, `inner_add_right`, `inner_sub_left`, `inner_sub_right`, `inner_neg_left`, `inner_neg_right`, `real_inner_smul_left`, `real_inner_smul_right`, `real_inner_comm`, `real_inner_self_nonneg`, and `inner_self_eq_zero`.

Lemma (reflection identities). ASSUME: V is a real inner product space and $\mathbf{r} \in V \setminus \{0\}$.
LET: $\tau_{\mathbf{r}}\mathbf{x} = \mathbf{x} - 2(\mathbf{r},\mathbf{x})\mathbf{r}/q(\mathbf{r})$.
PROVE: $\tau_{\mathbf{r}}$ is real linear, and, for all $\mathbf{x},\mathbf{y} \in V$,

$$\tau_{\mathbf{r}}^2\mathbf{x} = \mathbf{x}, \qquad (\tau_{\mathbf{r}}\mathbf{x},\tau_{\mathbf{r}}\mathbf{y}) = (\mathbf{x},\mathbf{y}), \qquad \tau_{\mathbf{r}}\mathbf{r} = -\mathbf{r}.$$

PROOF SKETCH: We will expand the reflection formula. This is the portion of Lemma (reflection identities) in `blueprint/sec3-remark311.tex` used here; the blueprint's s_α and (α,α) are $\tau_{\mathbf{r}}$ and $q(\mathbf{r})$ in the notation of this section.
⟨1⟩1. $q(\mathbf{r}) > 0$.
PROOF: Positive definiteness in Elementary identities and ⟨0⟩. □
⟨1⟩2. For $a,b \in \mathbb{R}$ and $\mathbf{x},\mathbf{y} \in V$, $\tau_{\mathbf{r}}(a\mathbf{x} + b\mathbf{y}) = a\tau_{\mathbf{r}}\mathbf{x} + b\tau_{\mathbf{r}}\mathbf{y}$.

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PROOF: Bilinearity in the second argument in the defining formula. □

⟨1⟩3. $(\mathbf{r}, \tau_{\mathbf{r}}\mathbf{x}) = -(\mathbf{r}, \mathbf{x})$ for every $\mathbf{x} \in V$.

PROOF: Bilinearity gives $(\mathbf{r}, \mathbf{x}) - 2(\mathbf{r}, \mathbf{x})q(\mathbf{r})/q(\mathbf{r})$; cancel using ⟨1⟩1. □

⟨1⟩4. $\tau_{\mathbf{r}}^2\mathbf{x} = \mathbf{x}$ for every $\mathbf{x} \in V$.

PROOF: Substitute ⟨1⟩3 in the second application of the defining formula: the two multiples of $\mathbf{r}$ cancel. □

⟨1⟩5. $(\tau_{\mathbf{r}}\mathbf{x}, \tau_{\mathbf{r}}\mathbf{y}) = (\mathbf{x}, \mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in V$.

PROOF: Bilinearity, symmetry, and ⟨1⟩1 give

$$(\tau_{\mathbf{r}}\mathbf{x}, \tau_{\mathbf{r}}\mathbf{y}) = (\mathbf{x}, \mathbf{y}) - \frac{4(\mathbf{r}, \mathbf{x})(\mathbf{r}, \mathbf{y})}{q(\mathbf{r})} + \frac{4(\mathbf{r}, \mathbf{x})(\mathbf{r}, \mathbf{y})}{q(\mathbf{r})} = (\mathbf{x}, \mathbf{y}).$$

□

⟨1⟩6. $\tau_{\mathbf{r}}\mathbf{r} = -\mathbf{r}$.

PROOF: The formula at $\mathbf{x} = \mathbf{r}$ is $\mathbf{r} - 2\mathbf{r}$, by ⟨1⟩1. □

⟨1⟩7. Q.E.D.

PROOF: Steps ⟨1⟩2, ⟨1⟩4, ⟨1⟩5, and ⟨1⟩6. □

1. Lines contained in a Root System

We will follow the ordinary proof in `cgss/proofs/thm312/thm312.tex`. The construction is the explicit set $\Psi(T)$ of Definition (the integer-span construction). All spaces are real inner product spaces; finite dimension is stated where needed. Entries of a vector or a matrix are function values. We will write $q(\mathbf{x}) = (\mathbf{x}, \mathbf{x})$ and $\mathbb{R}\mathbf{x} = \mathrm{span}_{\mathbb{R}}\{\mathbf{x}\}$. For a real scalar a, $a \in \mathbb{Z}$ and $a \in 2\mathbb{Z}$ mean $a = z$ and $a = 2z$, respectively, for some $z \in \mathbb{Z}$, with the integer cast into $\mathbb{R}$ understood.

The proof format is that of [L95]. Numbered assumptions of a result are cited as ⟨0⟩:1, and assumptions of an enclosing step as ⟨1⟩, ⟨2⟩, and so on. Each proof ends with a QED step; each leaf gives its facts. The tags **[new]** and **[repair]** identify new arguments and repairs to the printed paper. The metadata `\lean` and `\uses` record the Lean declarations and the dependencies.

1.1. Definitions and library facts.

Definition (angles and compatible colourings). ASSUME: V is a real inner product space, S is a set of real subspaces of V, and ℓ, m are real subspaces.

For $a \in \mathbb{R}$, let HasSquaredCosine(ℓ, m, a) mean

$$\forall \mathbf{x} \in \ell\ \forall \mathbf{y} \in m, \qquad (\mathbf{x}, \mathbf{y})^2 = a\, q(\mathbf{x})q(\mathbf{y}).$$

Let HasAllowedAngles(S) mean that for all distinct $\ell, m \in S$ there is $a \in \{0, 1/4, 1/2\}$ with this property. Let HasBipartition(S) mean that there is a function $b : S \to$ Bool such that, for all distinct $\ell, m \in S$,

$$\begin{aligned} b(\ell) = b(m) &\implies \text{HasSquaredCosine}(\ell, m, 0) \ \vee\ \text{HasSquaredCosine}(\ell, m, 1/4), \\ b(\ell) \neq b(m) &\implies \text{HasSquaredCosine}(\ell, m, 0) \ \vee\ \text{HasSquaredCosine}(\ell, m, 1/2). \end{aligned}$$

Neither colour fibre is required to be nonempty.

Definition (lines of a vector set). ASSUME: V is a real inner product space and $T \subseteq V$.

Define

$$\mathcal{L}(T) := \{\ell : \exists \mathbf{r} \in T,\ \ell = \mathbb{R}\mathbf{r}\}.$$

This is `LineSystem.rootLines`. When $0 \notin T$, each member has dimension one.

Definition (the integer-span construction). ASSUME: V is a real inner product space and $T \subseteq V$.

Put $L(T) = \operatorname{span}_{\mathbb{Z}}(T)$ and

$$\Psi(T) = \{\mathbf{x} \in L(T) : q(\mathbf{x}) = 2\} \cup \{\mathbf{x} \in L(T) : q(\mathbf{x}) = 4 \text{ and } (\mathbf{x}, \mathbf{y}) \in 2\mathbb{Z} \text{ for every } \mathbf{y} \in L(T)\}.$$

This is `LineSystem.integralSpanRoots`. The evenness condition quantifies over the entire fixed integer span.

Inner products and square roots. All library citations were checked in Mathlib at commit `082e2d37e8b0463410cdb532e111cd43d5a66174` (4 September 2026), the version pinned in `lake-manifest.json`. We will use Elementary identities, together with `real_inner_self_eq_norm_sq` and `abs_real_inner_le_norm` in `Mathlib/Analysis/InnerProductSpace/Basic.lean`. These give $(\mathbf{x}, \mathbf{y})^2 \le q(\mathbf{x})q(\mathbf{y})$. For positive real a, $\sqrt{a} > 0$ and $(\sqrt{a})^2 = a$ are `Real.sqrt_pos` and `Real.sq_sqrt` in `Mathlib/Analysis/Real/Sqrt.lean`.
Spans and finite-dimensional linear algebra. `finrank_eq_one_iff'` in `Mathlib/LinearAlgebra/Dimension/FreeAndStrongRankCondition.lean` gives a nonzero generator of a one-dimensional subspace. `eq_span_singleton_of_mem_of_finrank_eq_one` and `finrank_span_singleton` in `Mathlib/LinearAlgebra/FiniteDimensional/Basic.lean` identify its span and give the converse dimension assertion. `Submodule.mem_span_singleton` in `Mathlib/LinearAlgebra/Span/Defs.lean` expresses membership as a scalar multiple. `Submodule.subset_span`, `Submodule.span_mono`, and `Submodule.span_le` in the same file give membership of generators, monotonicity, and the criterion for a span to lie in a subspace. `Submodule.span_span_coe_preimage` there says that the preimage of a generating set under the inclusion of its span spans the whole subtype. `exists_linearIndependent` in `Mathlib/LinearAlgebra/LinearIndependent/Lemmas.lean` chooses an independent subset with the same real span. `LinearIndependent.finite` in `Mathlib/LinearAlgebra/Dimension/Finite.lean` makes that subset finite in finite dimension. `Submodule.span_induction` and `Submodule.span_induction₂` in `Mathlib/LinearAlgebra/Span/Defs.lean` give induction on a span, also over $\mathbb{Z}$. `Submodule.mem_span_range_iff_exists_fun` in `Mathlib/LinearAlgebra/Finsupp/LinearCombination.lean` expresses membership in a finite span by a finite sum. `Int.cast_smul_eq_zsmul` in `Mathlib/Algebra/Module/NatInt.lean` identifies multiplication by a real-cast integer with integer scalar multiplication.
Finite sets and cardinalities. `Set.Finite.of_injOn`, `Set.Finite.image`, `Set.Finite.subset`, and `Set.Finite.preimage` in `Mathlib/Data/Set/Finite/Basic.lean` give finiteness from an injection into a finite set, finiteness of images and subsets, and finiteness of a preimage when the map is injective on that preimage. A function set A^I is finite when A and I are finite.
Integer arithmetic and logarithms. Finite sums of integers are integers; finite sums of even integers are even. We will use field arithmetic, factorization of differences of squares, and integer inequalities in the arithmetic leaves. `zpow_add_one₀` in `Mathlib/Algebra/GroupWithZero/Basic.lean` gives the successor power identity for a nonzero base, including base 2. `Int.log` is the integer logarithm, and `Int.zpow_log_le_self`, `Int.lt_zpow_succ_log_self`, `Int.zpow_le_iff_le_log`, and `Int.lt_zpow_iff_log_lt` in `Mathlib/Data/Int/Log.lean` apply to base $2 > 1$ and $x > 0$. `Int.odd_iff` and `Int.not_even_iff_odd` in `Mathlib/Algebra/Ring/Int/Parity.lean` identify the two parities, also for negative integers. In particular $k + 1$ and $k - 1$ have parity opposite to k, by reduction modulo 2.
Orthogonal complements. `stdOrthonormalBasis` in `Mathlib/Analysis/InnerProductSpace/PiL2.lean` gives an orthonormal basis indexed by the finite dimension. `Submodule.inner_right_of_mem_orthogonal` in

`Mathlib/Analysis/InnerProductSpace/Orthogonal.lean` gives $(\mathbf{x}, \mathbf{y}) = 0$ for $\mathbf{x} \in W$ and $\mathbf{y} \in W^{\perp}$. `Orthonormal.inner_eq_zero` in `Mathlib/Analysis/InnerProductSpace/Orthonormal.lean` gives zero inner products at distinct indices of an orthonormal family. `Submodule.isCompl_orthogonal` and `Submodule.HasOrthogonalProjection.ofCompleteSpace` in `Mathlib/Analysis/InnerProductSpace/Projection/Basic.lean` give $W + W^{\perp} = V$. Here W is complete by the theorem `FiniteDimensional.complete` in `Mathlib/Topology/Algebra/Module/FiniteDimension.lean`.

These facts apply to the induced inner product on a subspace, including the zero subspace.

1.2. **Representatives and finiteness.**

Lemma (representatives of a line). ASSUME: V is a real inner product space, ℓ is a real subspace, $\mathbf{r} \in \ell \setminus \{0\}$, and $h > 0$.
PROVE: There is $\mathbf{t} \in \ell$ with $q(\mathbf{t}) = h$.

PROOF SKETCH: We will rescale the given nonzero vector.
⟨1⟩1. $q(\mathbf{r}) > 0$.
PROOF: Positive definiteness in the inner-product facts, ⟨0⟩. □
LET: $\mathbf{t} = (\sqrt{h}/\sqrt{q(\mathbf{r})})\mathbf{r}$.
⟨1⟩2. $\mathbf{t} \in \ell$.
PROOF: The subspace ℓ contains $\mathbf{r}$ by ⟨0⟩ and is closed under real scalar multiplication. □
⟨1⟩3. $q(\mathbf{t}) = h$.
PROOF: Bilinearity and the square-root facts give $q(\mathbf{t}) = h\,q(\mathbf{r})/q(\mathbf{r}) = h$, using $h > 0$ in ⟨0⟩ and ⟨1⟩1. □
⟨1⟩4. Q.E.D.
PROOF: The witness $\mathbf{t}$ satisfies ⟨1⟩2 and ⟨1⟩3. □

Lemma (squared cosine from representatives). ASSUME: 1. ℓ and m are subspaces of dimension one in the real inner product space V.
2. $\mathbf{r} \in \ell \setminus \{0\}$ and $\mathbf{s} \in m \setminus \{0\}$.
PROVE: For every $a \in \mathbb{R}$,

$$\text{HasSquaredCosine}(\ell, m, a) \iff (\mathbf{r}, \mathbf{s})^2 = a\,q(\mathbf{r})q(\mathbf{s}).$$

For all $a, a' \in \mathbb{R}$, if HasSquaredCosine(ℓ, m, a) and HasSquaredCosine(ℓ, m, a'), then $a = a'$.

PROOF SKETCH: We will write vectors on the two lines as scalar multiples of their representatives, then cancel their positive squared norms.
⟨1⟩1. $q(\mathbf{r})q(\mathbf{s}) > 0$.
PROOF: Positive definiteness and ⟨0⟩:2. □
⟨1⟩2. ASSUME: $a \in \mathbb{R}$ and HasSquaredCosine(ℓ, m, a).
PROVE: $(\mathbf{r}, \mathbf{s})^2 = a\,q(\mathbf{r})q(\mathbf{s})$.
PROOF: Substitute ⟨0⟩:2 in Definition (angles and compatible colourings). □
⟨1⟩3. ASSUME: $a \in \mathbb{R}$ and $(\mathbf{r}, \mathbf{s})^2 = a\,q(\mathbf{r})q(\mathbf{s})$.
PROVE: HasSquaredCosine(ℓ, m, a).
⟨2⟩1. For every $\mathbf{x} \in \ell$ and $\mathbf{y} \in m$ there are $u, v \in \mathbb{R}$ with $\mathbf{x} = u\mathbf{r}$ and $\mathbf{y} = v\mathbf{s}$.
PROOF: Apply the singleton-span facts to each nonzero representative in ⟨0⟩:2. □
⟨2⟩2. Q.E.D.
PROOF: Multiply ⟨1⟩ by u^2v^2 and use bilinearity with ⟨2⟩1; this includes $u = 0$ or $v = 0$. □
⟨1⟩4. ASSUME: $a, a' \in \mathbb{R}$, HasSquaredCosine(ℓ, m, a) and HasSquaredCosine(ℓ, m, a').
PROVE: $a = a'$.

PROOF: Apply ⟨1⟩2 at a and a'. The resulting equalities give $(a-a')q(\mathbf{r})q(\mathbf{s})=0$; cancel the positive product in ⟨1⟩1. □

⟨1⟩5. Q.E.D.

PROOF: Steps ⟨1⟩2 and ⟨1⟩3 give the equivalence; ⟨1⟩4 gives uniqueness. □

For lines, HasSquaredCosine(ℓ, m, a) with $a = 0, 1/4, 1/2$ means the unoriented angle is $90°, 60°, 45°$, respectively. We will use the division-free predicate and Lemma (squared cosine from representatives); no trigonometric fact is a premise below.

Lemma (inner-product coordinates). ASSUME: 1. V is a real inner product space, W a real subspace, and $(\mathbf{b}_i)_{i\in I}$ a family spanning W.
2. $\mathbf{x}, \mathbf{y} \in W$ and $(\mathbf{x}, \mathbf{b}_i) = (\mathbf{y}, \mathbf{b}_i)$ for every $i \in I$.
PROVE: $\mathbf{x} = \mathbf{y}$. Equivalently, $\mathbf{x} \mapsto ((\mathbf{x}, \mathbf{b}_i))_{i\in I}$ is injective on W.

PROOF SKETCH: We will show that the difference is orthogonal to its own span.

⟨1⟩1. $(\mathbf{x}-\mathbf{y}, \mathbf{b}_i) = 0$ for every $i \in I$.

PROOF: Subtract the equalities of ⟨0⟩:2 by bilinearity. □

⟨1⟩2. $(\mathbf{x}-\mathbf{y}, \mathbf{w}) = 0$ for every $\mathbf{w} \in W$.

PROOF: Real span induction from ⟨1⟩1: the assertion holds at zero and is preserved by addition and real scalar multiplication, by bilinearity. □

⟨1⟩3. $q(\mathbf{x}-\mathbf{y}) = 0$.

PROOF: Substitute $\mathbf{w} = \mathbf{x}-\mathbf{y} \in W$ in ⟨1⟩2. □

⟨1⟩4. Q.E.D.

PROOF: Positive definiteness applied to ⟨1⟩3. □

Lemma (finitely many lines). ASSUME: 1. V is a finite-dimensional real inner product space and S is a set of subspaces of dimension one.
2. HasAllowedAngles(S).
PROVE: S is finite.

PROOF SKETCH: We will encode all unit representatives by their inner products with a finite spanning family chosen from those representatives.

LET:
$$U = \{\mathbf{x} : \exists \ell \in S,\ \mathbf{x} \in \ell \text{ and } q(\mathbf{x}) = 1\}, \qquad W = \operatorname{span}_{\mathbb{R}} U,$$
$$\Theta = \{-1, -1/\sqrt{2}, -1/2, 0, 1/2, 1/\sqrt{2}, 1\}.$$

⟨1⟩1. PICK a finite family $(\mathbf{b}_i)_{i\in I}$ spanning W with every $\mathbf{b}_i \in U$.

PROOF: The independent-subset and finite-dimensional facts in Spans and finite-dimensional linear algebra, applied to U, give this family. The index set is empty if U is empty. □

⟨1⟩2. For every $\mathbf{x} \in U$ and $i \in I$, $(\mathbf{x}, \mathbf{b}_i) \in \Theta$.

⟨2⟩1. PICK $\ell, m \in S$ with $\mathbf{x} \in \ell$, $\mathbf{b}_i \in m$, and $q(\mathbf{x}) = q(\mathbf{b}_i) = 1$.

PROOF: The definition of U and ⟨1⟩1. □

LET: $p = (\mathbf{x}, \mathbf{b}_i)$.

⟨2⟩2. CASE: $\ell = m$.

PROVE: $p \in \{-1, 1\}$.

⟨3⟩1. There is $t \in \mathbb{R}$ with $\mathbf{x} = t\mathbf{b}_i$ and $t^2 = 1$.

PROOF: The singleton-span facts give the scalar; the two unit norms give its square by bilinearity. □

⟨3⟩2. Q.E.D.

PROOF: Factor $t^2 - 1$ in ⟨3⟩1; then $p = t \in \{-1, 1\}$. □

⟨2⟩3. CASE: $\ell \neq m$.

PROVE: $p \in \Theta$.

⟨3⟩1. $p^2 \in \{0, 1/4, 1/2\}$.
PROOF: Substitute the unit representatives in ⟨0⟩:2, using Definition (angles and compatible colourings). □
⟨3⟩2. CASE: $p^2 = 0$.
PROVE: $p = 0$.
PROOF: A real square is zero only at zero. □
⟨3⟩3. CASE: $p^2 = 1/4$.
PROVE: $p \in \{-1/2, 1/2\}$.
PROOF: Factor $(p - 1/2)(p + 1/2) = 0$. □
⟨3⟩4. CASE: $p^2 = 1/2$.
PROVE: $p \in \{-1/\sqrt{2}, 1/\sqrt{2}\}$.
PROOF: The square-root facts give $(1/\sqrt{2})^2 = 1/2$; factor $(p - 1/\sqrt{2})(p + 1/\sqrt{2}) = 0$. □
⟨3⟩5. Q.E.D.
PROOF: Step ⟨3⟩1 makes ⟨3⟩2, ⟨3⟩3, and ⟨3⟩4 exhaustive. □
⟨2⟩4. Q.E.D.
PROOF: The lines are equal or distinct; apply ⟨2⟩2 and ⟨2⟩3. □
⟨1⟩3. U is finite.
PROOF: By Lemma (inner-product coordinates) and ⟨1⟩1, the coordinate map is injective on $U \subseteq W$. Step ⟨1⟩2 puts its image in the finite set Θ^I; use the finite-set facts in Finite sets and cardinalities. □
⟨1⟩4. $\{\mathbb{R}\mathbf{x} : \mathbf{x} \in U\} = S$.
⟨2⟩1. For $\mathbf{x} \in U$, $\mathbb{R}\mathbf{x} \in S$.
PROOF: The witness line in the definition of U equals $\mathbb{R}\mathbf{x}$ by the singleton-span facts, since $q(\mathbf{x}) = 1$. □
⟨2⟩2. For $\ell \in S$, some $\mathbf{x} \in U$ satisfies $\ell = \mathbb{R}\mathbf{x}$.
PROOF: A nonzero generator exists by the dimension-one facts. Lemma (representatives of a line), with $h = 1$, normalizes it; the singleton-span facts identify its line. □
⟨2⟩3. Q.E.D.
PROOF: The two inclusions are ⟨2⟩1 and ⟨2⟩2. □
⟨1⟩5. Q.E.D.
PROOF: The image of the finite set in ⟨1⟩3 is finite; use ⟨1⟩4. □

Lemma (a bounded part of an integer span). ASSUME: 1. V is a finite-dimensional real inner product space and $T \subseteq V$ is finite.
2. $q(\mathbf{t}) \in \{2, 4\}$ for $\mathbf{t} \in T$, and $(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$ for $\mathbf{t}, \mathbf{u} \in T$.
LET: $L = \operatorname{span}_{\mathbb{Z}} T$, $W = \operatorname{span}_{\mathbb{R}} T$, $B = \{\mathbf{x} \in L : q(\mathbf{x}) \leq 4\}$.
PROVE: 1. $L \subseteq W$, and $(\mathbf{x}, \mathbf{y}) \in \mathbb{Z}$ for all $\mathbf{x}, \mathbf{y} \in L$.
2. For every $\mathbf{t} \in T$ such that $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ for all $\mathbf{u} \in T$, we have $(\mathbf{t}, \mathbf{x}) \in 2\mathbb{Z}$ for all $\mathbf{x} \in L$.
3. B is finite.

PROOF SKETCH: [**new**] We will expand integer combinations and use bounded inner-product coordinates against T itself. This simplifies the ordinary draft's basis argument; no independence of T is assumed.
⟨1⟩1. Every $\mathbf{x} \in L$ has an expression $\mathbf{x} = \sum_{\mathbf{t} \in T} a(\mathbf{t})\mathbf{t}$ with $a : T \to \mathbb{Z}$.
PROOF: Apply the finite-span formula in Spans and finite-dimensional linear algebra over $\mathbb{Z}$ to the inclusion $T \to V$. □
⟨1⟩2. $L \subseteq W$.
PROOF: Cast the integer coefficients in ⟨1⟩1 to real coefficients. □
⟨1⟩3. For all $\mathbf{x}, \mathbf{y} \in L$, $(\mathbf{x}, \mathbf{y}) \in \mathbb{Z}$.

⟨2⟩1. For coefficients $a, b : T \to \mathbb{Z}$,
$$\left(\sum_{\mathbf{t}\in T} a(\mathbf{t})\mathbf{t}, \sum_{\mathbf{u}\in T} b(\mathbf{u})\mathbf{u}\right) = \sum_{\mathbf{t},\mathbf{u}\in T} a(\mathbf{t})b(\mathbf{u})(\mathbf{t},\mathbf{u}).$$
PROOF: Bilinearity, iterated over the two finite sums. □
⟨2⟩2. Q.E.D.
PROOF: Each summand in ⟨2⟩1 is an integer by ⟨0⟩:2; use ⟨1⟩1 and closure under finite sums. □
⟨1⟩4. ASSUME: $\mathbf{t} \in T$ and $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ for every $\mathbf{u} \in T$.
PROVE: $(\mathbf{t}, \mathbf{x}) \in 2\mathbb{Z}$ for every $\mathbf{x} \in L$.
PROOF: Bilinearity and ⟨1⟩1 give $(\mathbf{t}, \mathbf{x}) = \sum_{\mathbf{u}\in T} a(\mathbf{u})(\mathbf{t}, \mathbf{u})$; every summand is even by ⟨1⟩, so the sum is even. □
⟨1⟩5. For $\mathbf{x} \in B$ and $\mathbf{t} \in T$, $(\mathbf{x}, \mathbf{t}) \in \{-4, -3, \ldots, 4\}$.
⟨2⟩1. $(\mathbf{x}, \mathbf{t})^2 \le 16$.
PROOF: Cauchy–Schwarz in Inner products and square roots gives $(\mathbf{x}, \mathbf{t})^2 \le q(\mathbf{x})q(\mathbf{t}) \le 4 \cdot 4$, by the definition of B, $\mathbf{t} \in T$, and ⟨0⟩:2. □
⟨2⟩2. $(\mathbf{x}, \mathbf{t}) \in \mathbb{Z}$.
PROOF: Step ⟨1⟩3, since $\mathbf{x} \in L$ and $\mathbf{t} \in T \subseteq L$ by `Submodule.subset_span`. □
⟨2⟩3. Q.E.D.
PROOF: An integer whose square is at most 16 lies between -4 and 4; apply ⟨2⟩1 and ⟨2⟩2. □
⟨1⟩6. B is finite.
PROOF: The family indexed by T spans W by definition. Lemma (inner-product coordinates) makes $\mathbf{x} \mapsto ((\mathbf{x}, \mathbf{t}))_{\mathbf{t}\in T}$ injective on W, which contains B by ⟨1⟩2. Step ⟨1⟩5 puts its image in the finite function set $\{-4, -3, \ldots, 4\}^T$, since T is finite by ⟨0⟩:1. Apply `Set.Finite.of_injOn`. □
⟨1⟩7. Q.E.D.
PROOF: Steps ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, and ⟨1⟩6. For $T = \varnothing$ the same proof uses the empty coordinate family and gives $B = \{0\}$. □

1.3. **The explicit construction.**

Lemma (reflection adjointness). ASSUME: V is a real inner product space and $\mathbf{r} \in V \setminus \{0\}$.
PROVE: For all $\mathbf{x}, \mathbf{y} \in V$, $(\tau_{\mathbf{r}}\mathbf{x}, \mathbf{y}) = (\mathbf{x}, \tau_{\mathbf{r}}\mathbf{y})$.

PROOF SKETCH: We will use the metric identity and involution in Lemma (reflection identities).
⟨1⟩1. $(\tau_{\mathbf{r}}\mathbf{x}, \mathbf{y}) = (\mathbf{x}, \tau_{\mathbf{r}}\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in V$.
PROOF: Apply the metric identity in Lemma (reflection identities) to $\mathbf{x}, \tau_{\mathbf{r}}\mathbf{y}$; its involution identity changes the left side to $(\tau_{\mathbf{r}}\mathbf{x}, \mathbf{y})$. Its nonzero hypothesis is ⟨0⟩. □
⟨1⟩2. Q.E.D.
PROOF: Step ⟨1⟩1. □

Lemma (integral proportional vectors). ASSUME: 1. V is a real inner product space, $\mathbf{r}, \mathbf{s} \in V$, and $q(\mathbf{r}), q(\mathbf{s}) \in \{2, 4\}$.
2. $(\mathbf{r}, \mathbf{s}) \in \mathbb{Z}$ and $\mathbf{s} = t\mathbf{r}$ for $t \in \mathbb{R}$.
PROVE: $t = 1$ or $t = -1$.

PROOF SKETCH: **[new]** We will use $t^2 = 1$ for equal lengths. For unequal lengths the integer inner product would have square 8, simplifying the ordinary draft's irrational-square-root argument.
⟨1⟩1. $q(\mathbf{s}) = t^2 q(\mathbf{r})$.
PROOF: Bilinearity and $\mathbf{s} = t\mathbf{r}$ in ⟨0⟩:2. □
⟨1⟩2. CASE: $q(\mathbf{r}) = q(\mathbf{s})$.
PROVE: $t \in \{1, -1\}$.

PROOF: Divide ⟨1⟩1 by $q(\mathbf{r}) > 0$ and factor $t^2 - 1$. □
⟨1⟩3. $(\mathbf{r}, \mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$.
PROOF: Bilinearity gives $(\mathbf{r}, \mathbf{s}) = tq(\mathbf{r})$. Square this equality and substitute ⟨1⟩1. □
⟨1⟩4. CASE: $q(\mathbf{r}) \neq q(\mathbf{s})$.
PROVE: False.
⟨2⟩1. $(\mathbf{r}, \mathbf{s})^2 = 8$.
PROOF: The two unequal members of $\{2, 4\}$ in ⟨0⟩:1 have product 8; substitute in ⟨1⟩3. □
⟨2⟩2. PICK $z \in \mathbb{Z}$ with $(\mathbf{r}, \mathbf{s}) = z$.
PROOF: Integrality in ⟨0⟩:2. □
⟨2⟩3. Q.E.D.
PROOF: Steps ⟨2⟩1 and ⟨2⟩2 give $z^2 = 8$. Thus $4 < |z|^2 < 9$, so $2 < |z| < 3$, impossible for an integer. □
⟨1⟩5. Q.E.D.
PROOF: Equality or inequality of the squared lengths gives ⟨1⟩2 or ⟨1⟩4. The unequal-length case is impossible. □

Proposition (the integer-span Root System). ASSUME: 1. V is a finite-dimensional real inner product space, $T \subseteq V$ is finite, and $\operatorname{span}_{\mathbb{R}} T = V$.
2. $q(\mathbf{t}) \in \{2, 4\}$ for every $\mathbf{t} \in T$.
3. $(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$ for every $\mathbf{t}, \mathbf{u} \in T$.
4. If $\mathbf{t} \in T$ and $q(\mathbf{t}) = 4$, then $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ for every $\mathbf{u} \in T$.
PROVE: IsRootSet($\Psi(T)$), $T \subseteq \Psi(T)$, and $q(\mathbf{r}) \in \{2, 4\}$ for every $\mathbf{r} \in \Psi(T)$.

PROOF SKETCH: **[repair]** This fills part of Gap 11. The explicit set in Definition (the integer-span construction) replaces a reflection closure, which would need stages, a union and a leastness argument. We will check each Root System requirement, keeping the evenness condition invariant under reflection.
LET: $L = \operatorname{span}_{\mathbb{Z}} T$ and $\Psi = \Psi(T)$.
⟨1⟩1. $\Psi \subseteq L$, every $\mathbf{r} \in \Psi$ has $q(\mathbf{r}) \in \{2, 4\}$, and $0 \notin \Psi$.
PROOF: Definition (the integer-span construction) and $q(0) = 0$. □
⟨1⟩2. $T \subseteq \Psi$.
⟨2⟩1. ASSUME: $\mathbf{t} \in T$ and $q(\mathbf{t}) = 2$.
PROVE: $\mathbf{t} \in \Psi$.
PROOF: By `Submodule.subset_span`, $T \subseteq L$. The membership and length in ⟨2⟩ give the first clause of Definition (the integer-span construction). □
⟨2⟩2. ASSUME: $\mathbf{t} \in T$ and $q(\mathbf{t}) = 4$.
PROVE: $\mathbf{t} \in \Psi$.
⟨3⟩1. $(\mathbf{t}, \mathbf{x}) \in 2\mathbb{Z}$ for every $\mathbf{x} \in L$.
PROOF: Lemma (a bounded part of an integer span), part 2, applied with ⟨0⟩:1, ⟨0⟩:2, and ⟨0⟩:3; the vector belongs to T and has squared norm 4 by ⟨2⟩, so ⟨0⟩:4 supplies its even pairings with T. □
⟨3⟩2. Q.E.D.
PROOF: The second clause of Definition (the integer-span construction) follows from $\mathbf{t} \in L$, the case length, and ⟨3⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Hypothesis ⟨0⟩:2 makes ⟨2⟩1 and ⟨2⟩2 exhaustive. □
⟨1⟩3. Ψ is finite.
PROOF: By ⟨1⟩1, it is contained in $\{\mathbf{x} \in L : q(\mathbf{x}) \leq 4\}$, finite by part 3 of Lemma (a bounded part of an integer span) with ⟨0⟩:1, ⟨0⟩:2, and ⟨0⟩:3. Apply `Set.Finite.subset`. □
⟨1⟩4. $\operatorname{span}_{\mathbb{R}} \Psi = V$.

PROOF: `Submodule.span_mono`, ⟨1⟩2, and ⟨0⟩:1. □

⟨1⟩5. For $\mathbf{r} \in \Psi$ and $\mathbf{x} \in L$, $2(\mathbf{r}, \mathbf{x})/q(\mathbf{r}) \in \mathbb{Z}$.

⟨2⟩1. CASE: $q(\mathbf{r}) = 2$.

PROOF: The coefficient is $(\mathbf{r}, \mathbf{x})$, an integer by Lemma (a bounded part of an integer span) and ⟨1⟩1. □

⟨2⟩2. CASE: $q(\mathbf{r}) = 4$.

PROOF: The coefficient is $(\mathbf{r}, \mathbf{x})/2$, an integer by the evenness clause of Definition (the integer-span construction). □

⟨2⟩3. Q.E.D.

PROOF: Step ⟨1⟩1 makes ⟨2⟩1 and ⟨2⟩2 exhaustive. □

⟨1⟩6. For $\mathbf{r} \in \Psi$ and $\mathbf{x} \in L$, $\tau_{\mathbf{r}}\mathbf{x} \in L$.

PROOF: Step ⟨1⟩5 gives an integer z equal to the reflection coefficient. By `Int.cast_smul_eq_zsmul`, the term $(z : \mathbb{R})\mathbf{r}$ is the integer multiple $z\mathbf{r}$. Step ⟨1⟩1 puts $\mathbf{r}$ in L; the integer submodule L is closed under integer multiples and subtraction. □

⟨1⟩7. For $\mathbf{r}, \mathbf{s} \in \Psi$, $\tau_{\mathbf{r}}\mathbf{s} \in \Psi$.

LET: $\mathbf{u} = \tau_{\mathbf{r}}\mathbf{s}$.

⟨2⟩1. $\mathbf{u} \in L$ and $q(\mathbf{u}) = q(\mathbf{s})$.

PROOF: Step ⟨1⟩6 and the metric identity of Lemma (reflection identities) at $\mathbf{s}, \mathbf{s}$, using the nonzero assertion in ⟨1⟩1. □

⟨2⟩2. CASE: $q(\mathbf{s}) = 2$.

PROVE: $\mathbf{u} \in \Psi$.

PROOF: The first clause of Definition (the integer-span construction), with ⟨2⟩1. □

⟨2⟩3. CASE: $q(\mathbf{s}) = 4$.

PROVE: $\mathbf{u} \in \Psi$.

⟨3⟩1. For every $\mathbf{x} \in L$, $(\mathbf{u}, \mathbf{x}) \in 2\mathbb{Z}$.

PROOF: Lemma (reflection adjointness) gives $(\mathbf{u}, \mathbf{x}) = (\mathbf{s}, \tau_{\mathbf{r}}\mathbf{x})$. This is even by the long-root clause for $\mathbf{s}$ and ⟨1⟩6. □

⟨3⟩2. Q.E.D.

PROOF: Apply the second clause of Definition (the integer-span construction) to ⟨2⟩1 and ⟨3⟩1. □

⟨2⟩4. Q.E.D.

PROOF: Step ⟨1⟩1 makes ⟨2⟩2 and ⟨2⟩3 exhaustive. □

⟨1⟩8. For $\mathbf{r}, t\mathbf{r} \in \Psi$, $t \in \{1, -1\}$.

PROOF: Apply Lemma (integral proportional vectors); its length hypotheses are ⟨1⟩1, and its inner-product hypothesis is part 1 of Lemma (a bounded part of an integer span), applied with ⟨0⟩:1, ⟨0⟩:2, and ⟨0⟩:3 and the inclusion $\Psi \subseteq L$ from ⟨1⟩1. □

⟨1⟩9. Q.E.D.

PROOF: Definition 3.9 has six requirements: ⟨1⟩3, ⟨1⟩1, ⟨1⟩4, ⟨1⟩5 restricted to $\Psi \subseteq L$, ⟨1⟩8, and ⟨1⟩7 supply them. The additional conclusions are ⟨1⟩2 and ⟨1⟩1. □

Lemma (completion of the seeds). ASSUME: 1. V is a finite-dimensional real inner product space and $T \subseteq V$ is finite.
2. $q(\mathbf{t}) \in \{2, 4\}$ for every $\mathbf{t} \in T$.
3. $(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$ for every $\mathbf{t}, \mathbf{u} \in T$.
4. If $\mathbf{t} \in T$ and $q(\mathbf{t}) = 4$, then $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ for every $\mathbf{u} \in T$.

PROVE: There is a finite set $T' \supseteq T$ spanning V with $q(\mathbf{t}) \in \{2, 4\}$ for $\mathbf{t} \in T'$, all inner products in T' integral, and $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ whenever $\mathbf{t}, \mathbf{u} \in T'$ and $q(\mathbf{t}) = 4$.

PROOF SKETCH: We will add scaled orthonormal vectors of the orthogonal complement. The diagonal inner product of a new vector is 2.

LET: $W = \operatorname{span}_{\mathbb{R}} T$.
⟨1⟩1. PICK a finite orthonormal basis $(\mathbf{e}_i)_{i \in I}$ of $W^{\perp}$, viewed in V.
PROOF: The orthogonal-complement facts in Orthogonal complements. □
LET: $\mathbf{u}_i = \sqrt{2}\mathbf{e}_i$, $E = \{\mathbf{u}_i : i \in I\}$, and $T' = T \cup E$.
⟨1⟩2. $q(\mathbf{u}_i) = 2$ for every $i \in I$.
PROOF: Orthonormality in ⟨1⟩1 and bilinearity with $(\sqrt{2})^2 = 2$. □
⟨1⟩3. All inner products in T' are integers.
⟨2⟩1. ASSUME: $\mathbf{t}, \mathbf{u} \in T$.
PROVE: $(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$.
PROOF: Hypothesis ⟨0⟩:3. □
⟨2⟩2. ASSUME: $\mathbf{t} \in T$ and $i \in I$.
PROVE: $(\mathbf{t}, \mathbf{u}_i) = 0$.
PROOF: Here $\mathbf{t} \in W$ and $\mathbf{e}_i \in W^{\perp}$ by ⟨1⟩1; `Submodule.subset_span` gives the first membership. Apply `Submodule.inner_right_of_mem_orthogonal` and multiply the zero inner product by $\sqrt{2}$. □
⟨2⟩3. ASSUME: $\mathbf{t} \in T$ and $i \in I$.
PROVE: $(\mathbf{u}_i, \mathbf{t}) = 0$.
PROOF: Symmetry and ⟨2⟩2. □
⟨2⟩4. ASSUME: $i \in I$.
PROVE: $(\mathbf{u}_i, \mathbf{u}_i) = 2$.
PROOF: Step ⟨1⟩2. □
⟨2⟩5. ASSUME: $i, j \in I$ and $i \neq j$.
PROVE: $(\mathbf{u}_i, \mathbf{u}_j) = 0$.
PROOF: Apply `Orthonormal.inner_eq_zero` to ⟨1⟩1 and $i \neq j$ in ⟨2⟩; bilinearity preserves zero after scaling. □
⟨2⟩6. Q.E.D.
PROOF: Membership in $T \cup E$ gives the old/old, old/new, new/old, and new/new possibilities. The last splits into equal and distinct indices. The preceding five steps give integers in every case. □
⟨1⟩4. For $\mathbf{t}, \mathbf{u} \in T'$ with $q(\mathbf{t}) = 4$, $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$.
⟨2⟩1. $\mathbf{t} \in T$.
PROOF: Step ⟨1⟩2 excludes $\mathbf{t} \in E$; use $\mathbf{t} \in T \cup E$. □
⟨2⟩2. CASE: $\mathbf{u} \in T$.
PROOF: Hypothesis ⟨0⟩:4 and ⟨2⟩1. □
⟨2⟩3. CASE: $\mathbf{u} \in E$.
PROOF: Step ⟨2⟩1 and `Submodule.subset_span` put $\mathbf{t}$ in W. The definition of E puts $\mathbf{u}$ in $W^{\perp}$. The inner product is zero by `Submodule.inner_right_of_mem_orthogonal`. □
⟨2⟩4. Q.E.D.
PROOF: The two cases cover $\mathbf{u} \in T \cup E$. □
⟨1⟩5. $\operatorname{span}_{\mathbb{R}} T' = V$.
⟨2⟩1. $W \subseteq \operatorname{span}_{\mathbb{R}} T'$.
PROOF: `Submodule.span_mono` and $T \subseteq T'$. □
⟨2⟩2. $W^{\perp} \subseteq \operatorname{span}_{\mathbb{R}} T'$.
PROOF: Since $\sqrt{2} \neq 0$, each basis vector is $\mathbf{e}_i = (1/\sqrt{2})\mathbf{u}_i \in \operatorname{span}_{\mathbb{R}} T'$. The basis in ⟨1⟩1 spans $W^{\perp}$. □
⟨2⟩3. Q.E.D.
PROOF: The orthogonal decomposition $W + W^{\perp} = V$, with ⟨2⟩1 and ⟨2⟩2. □
⟨1⟩6. Q.E.D.

PROOF: The set E is a finite image of I, so T' is finite and contains T. The lengths follow from $\langle 0\rangle$:2 and $\langle 1\rangle 2$; the other properties are $\langle 1\rangle 3$, $\langle 1\rangle 4$, and $\langle 1\rangle 5$. The construction allows an empty basis. □

Lemma (seeds from a colouring). ASSUME: 1. V is a finite-dimensional real inner product space, and S is a set of subspaces of dimension one.
2. HasBipartition(S).
LET: $W = \operatorname{span}_{\mathbb{R}}(\bigcup_{\ell\in S} \ell)$.
PROVE: There is a finite set $T \subseteq V$ such that
1. $\mathcal{L}(T) = S$ and $\operatorname{span}_{\mathbb{R}} T = W$;
2. $q(\mathbf{t}) \in \{2, 4\}$ for every $\mathbf{t} \in T$;
3. $(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$ for every $\mathbf{t}, \mathbf{u} \in T$; and
4. $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ whenever $\mathbf{t}, \mathbf{u} \in T$ and $q(\mathbf{t}) = 4$.

PROOF SKETCH: We will give the two colour classes squared lengths 2 and 4 and check each ordered pair of colours.
$\langle 1\rangle 1$. PICK a compatible colouring $b : S \to$ Bool.
PROOF: Hypothesis $\langle 0\rangle$:2 and Definition (angles and compatible colourings). □
$\langle 1\rangle 2$. HasAllowedAngles(S) and S is finite.
$\langle 2\rangle 1$. ASSUME: $\ell, m \in S$, $\ell \neq m$ and $b(\ell) = b(m)$.
PROVE: There is $a \in \{0, 1/4, 1/2\}$ with HasSquaredCosine(ℓ, m, a).
PROOF: The equal-colour clause of $\langle 1\rangle 1$ gives a parameter in $\{0, 1/4\}$ with the required predicate. □
$\langle 2\rangle 2$. ASSUME: $\ell, m \in S$, $\ell \neq m$ and $b(\ell) \neq b(m)$.
PROVE: There is $a \in \{0, 1/4, 1/2\}$ with HasSquaredCosine(ℓ, m, a).
PROOF: The different-colour clause of $\langle 1\rangle 1$ gives a parameter in $\{0, 1/2\}$ with the required predicate. □
$\langle 2\rangle 3$. Q.E.D.
PROOF: Equality or inequality of the colours exhausts all pairs. Definition (angles and compatible colourings) gives allowed angles, and Lemma (finitely many lines) gives finiteness. □
$\langle 1\rangle 3$. PICK $\mathbf{t}_\ell \in \ell$ for every $\ell \in S$, with $q(\mathbf{t}_\ell) = 2$ when $b(\ell) =$ false and $q(\mathbf{t}_\ell) = 4$ when $b(\ell) =$ true.
PROOF: Choose a nonzero generator by the dimension-one facts, then apply Lemma (representatives of a line) with $h = 2$ or $h = 4$. Choice on the empty set gives the empty family. □
LET: $T = \{\mathbf{t}_\ell : \ell \in S\}$.
$\langle 1\rangle 4$. T is finite, $\mathcal{L}(T) = S$, and $\operatorname{span}_{\mathbb{R}} T = W$.
$\langle 2\rangle 1$. T is finite.
PROOF: It is an image of the finite set in $\langle 1\rangle 2$. □
$\langle 2\rangle 2$. $\mathbb{R}\mathbf{t}_\ell = \ell$ for every $\ell \in S$.
PROOF: Step $\langle 1\rangle 3$ gives a nonzero member; use the singleton-span facts in Spans and finite-dimensional linear algebra. □
$\langle 2\rangle 3$. $\operatorname{span}_{\mathbb{R}} T \subseteq W$.
PROOF: Every seed belongs to its line in S by $\langle 1\rangle 3$; use the definition of W and `Submodule.span_mono`. □
$\langle 2\rangle 4$. $W \subseteq \operatorname{span}_{\mathbb{R}} T$.
PROOF: Step $\langle 2\rangle 2$ puts every line in S in the span of T, by `Submodule.subset_span` and `Submodule.span_le`. Apply `Submodule.span_le` again to their union. □
$\langle 2\rangle 5$. Q.E.D.
PROOF: Definition (lines of a vector set) and $\langle 2\rangle 2$ give $\mathcal{L}(T) = S$. The other assertions are $\langle 2\rangle 1$, $\langle 2\rangle 3$, and $\langle 2\rangle 4$. □
$\langle 1\rangle 5$. For $\ell, m \in S$, $(\mathbf{t}_\ell, \mathbf{t}_m)$ is an integer, and is even if $q(\mathbf{t}_\ell) = 4$.

LET: $p = (\mathbf{t}_\ell, \mathbf{t}_m)$.
⟨2⟩1. CASE: $\ell = m$.
PROOF: Step ⟨1⟩3 gives $p = 2$ or $p = 4$; both are even integers. □
⟨2⟩2. CASE: $\ell \neq m$ and $b(\ell) = b(m) = \text{false}$.
⟨3⟩1. $p^2 \in \{0, 1\}$.
PROOF: The norms in ⟨1⟩3 have product 4; substitute them in the equal-colour clause of ⟨1⟩1. □
⟨3⟩2. Q.E.D.
PROOF: Factoring the two possible square equations in ⟨3⟩1 gives $p \in \{0, 1, -1\}$. The evenness implication has false hypothesis, since $q(\mathbf{t}_\ell) = 2$. □
⟨2⟩3. CASE: $\ell \neq m$ and $b(\ell) = b(m) = \text{true}$.
⟨3⟩1. $p^2 \in \{0, 4\}$.
PROOF: The norms in ⟨1⟩3 have product 16; use the equal-colour clause of ⟨1⟩1. □
⟨3⟩2. Q.E.D.
PROOF: Factoring the square equations in ⟨3⟩1 gives $p \in \{0, 2, -2\}$, all even integers. □
⟨2⟩4. CASE: $\ell \neq m$, $b(\ell) = \text{false}$ and $b(m) = \text{true}$.
⟨3⟩1. $p^2 \in \{0, 4\}$.
PROOF: The norms have product 8 by ⟨1⟩3; use the different-colour clause of ⟨1⟩1. □
⟨3⟩2. Q.E.D.
PROOF: Factoring ⟨3⟩1 gives $p \in \{0, 2, -2\}$, all even integers. □
⟨2⟩5. CASE: $\ell \neq m$, $b(\ell) = \text{true}$ and $b(m) = \text{false}$.
⟨3⟩1. $p^2 \in \{0, 4\}$.
PROOF: The norms have product 8 by ⟨1⟩3; use the different-colour clause of ⟨1⟩1. □
⟨3⟩2. Q.E.D.
PROOF: Factoring ⟨3⟩1 gives $p \in \{0, 2, -2\}$, all even integers. □
⟨2⟩6. Q.E.D.
PROOF: The diagonal and the four ordered Boolean pairs exhaust the possibilities, as in ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4, and ⟨2⟩5. □
⟨1⟩6. Q.E.D.
PROOF: Step ⟨1⟩4 supplies part 1 and finiteness; ⟨1⟩3 supplies part 2; and ⟨1⟩5 supplies parts 3 and 4 on the image T. □

Corollary (normalized containment). ASSUME: 1. V is a finite-dimensional real inner product space, and S is a set of subspaces of dimension one.
2. HasBipartition(S).
LET: $W = \operatorname{span}_{\mathbb{R}}(\bigcup_{\ell \in S} \ell)$, and let $\iota : W \to V$ be the inclusion.
PROVE: 1. There is $\Phi \subseteq V$ with IsRootSet(Φ), $S \subseteq \mathcal{L}(\Phi)$, and $q(\mathbf{r}) \in \{2, 4\}$ for $\mathbf{r} \in \Phi$.
2. There is $\Phi_W \subseteq W$ with IsRootSet(Φ_W) in the induced inner product on W, $S \subseteq \mathcal{L}(\iota(\Phi_W))$, and $q(\mathbf{r}) \in \{2, 4\}$ for $\mathbf{r} \in \Phi_W$.

PROOF SKETCH: We will apply the construction in V after completing seeds. For the span assertion we will apply the same proposition to the seeds regarded as vectors in the separate ambient type W.
⟨1⟩1. PICK a finite set T such that $\mathcal{L}(T) = S$, $\operatorname{span}_{\mathbb{R}} T = W$, each squared length in T is 2 or 4, all inner products in T are integers, and each length-4 vector in T pairs evenly with every vector in T.
PROOF: Lemma (seeds from a colouring) applies to ⟨0⟩:1 and ⟨0⟩:2. □
⟨1⟩2. There is a set Φ satisfying part 1.
⟨2⟩1. PICK a finite $T' \supseteq T$ spanning V, with every squared length 2 or 4, all inner products integral, and every length-4 vector pairing evenly with all of T'.

PROOF: Apply Lemma (completion of the seeds) to the conditions in ⟨1⟩1 and the finite dimension in ⟨0⟩:1. □

⟨2⟩2. $\Phi = \Psi(T')$ is a Root System in V, contains T', and has squared lengths 2 or 4.
PROOF: Proposition (the integer-span Root System) with ⟨2⟩1. □

⟨2⟩3. Q.E.D.
PROOF: Definition (lines of a vector set) sends $T \subseteq T' \subseteq \Phi$ to $S = \mathcal{L}(T) \subseteq \mathcal{L}(\Phi)$, by ⟨1⟩1 and ⟨2⟩2. □

⟨1⟩3. There is a set Φ_W satisfying part 2.
LET: $T_W = \{\mathbf{w} \in W : \iota\mathbf{w} \in T\}$.

⟨2⟩1. T_W is finite and $\iota(T_W) = T$.
PROOF: The inclusion is injective, and every member of T lies in W by ⟨1⟩1; it restricts to a bijection $T_W \to T$. Finiteness follows from `Set.Finite.preimage`, applied to the finite T and the injectivity of ι on T_W. □

⟨2⟩2. $\operatorname{span}_{\mathbb{R}} T_W = W$ as a subspace of W.
PROOF: Rewrite $W = \operatorname{span}_{\mathbb{R}} T$ using ⟨1⟩1. The assertion is `Submodule.span_span_coe_preimage` for the inclusion ι. □

⟨2⟩3. Every squared length in T_W is 2 or 4, all its inner products are integers, and each length-4 vector pairs evenly with all of T_W.
PROOF: The induced inner product on W is the ambient inner product; substitute ι in the three conditions for T in ⟨1⟩1. □

⟨2⟩4. $\Phi_W = \Psi(T_W)$ is a Root System in W, contains T_W, and has squared lengths 2 or 4.
PROOF: Apply Proposition (the integer-span Root System) in the finite-dimensional inner product space W, with ⟨2⟩1, ⟨2⟩2, and ⟨2⟩3. □

⟨2⟩5. Q.E.D.
PROOF: By ⟨2⟩1 and ⟨2⟩4, $T \subseteq \iota(\Phi_W)$. Definition (lines of a vector set) and ⟨1⟩1 give $S \subseteq \mathcal{L}(\iota(\Phi_W))$. □

⟨1⟩4. Q.E.D.
PROOF: Steps ⟨1⟩2 and ⟨1⟩3. □

1.4. **Lengths and the necessary colouring.**

Lemma (lengths on root lines). ASSUME: 1. V is a finite-dimensional real inner product space and IsRootSet(Φ).
2. $\mathbf{r}, \mathbf{s} \in \Phi$.
PROVE: 1. If $4(\mathbf{r},\mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$, then $q(\mathbf{r}) = q(\mathbf{s})$.
2. If $2(\mathbf{r},\mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$, then $q(\mathbf{s}) = 2q(\mathbf{r})$ or $q(\mathbf{r}) = 2q(\mathbf{s})$.

PROOF SKETCH: We will use the two ordered crystallographic coefficients. This lemma keeps the integrality argument separate from *angles and fixed-length subsystems* (`lem:angles`) in the blueprint, whose later conclusions use Theorem B. Here neither reducedness nor classification is needed.

⟨1⟩1. $q(\mathbf{r}), q(\mathbf{s}) > 0$.
PROOF: Nonzero roots in Definition 3.9, ⟨0⟩:2, and positive definiteness in Inner products and square roots. □

LET: $u = 2(\mathbf{r},\mathbf{s})/q(\mathbf{r})$ and $v = 2(\mathbf{r},\mathbf{s})/q(\mathbf{s})$.

⟨1⟩2. ASSUME: $(\mathbf{r},\mathbf{s}) \neq 0$.
PROVE: u, v are nonzero integers of the same sign, and

$$|u||v| = uv = \frac{4(\mathbf{r},\mathbf{s})^2}{q(\mathbf{r})q(\mathbf{s})}, \qquad \frac{|u|}{|v|} = \frac{u}{v} = \frac{q(\mathbf{s})}{q(\mathbf{r})}.$$

⟨2⟩1. $u, v \in \mathbb{Z}$.

PROOF: Definition 3.9 for the two ordered pairs in ⟨0⟩:2, using symmetry for v. □

⟨2⟩2. u, v have the same nonzero sign.
PROOF: Their common numerator is nonzero by ⟨1⟩, and their denominators are positive by ⟨1⟩1. □

⟨2⟩3. $|u||v| = uv = 4(\mathbf{r},\mathbf{s})^2/(q(\mathbf{r})q(\mathbf{s}))$.
PROOF: Multiply the defining fractions; the equal signs in ⟨2⟩2 give the absolute-value identity. □

⟨2⟩4. Q.E.D.
PROOF: Cancel the common nonzero numerator to obtain $u/v = q(\mathbf{s})/q(\mathbf{r})$. Step ⟨2⟩2 gives $|u|/|v| = u/v$; combine with ⟨2⟩1 and ⟨2⟩3. □

⟨1⟩3. ASSUME: $4(\mathbf{r},\mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$.
PROVE: $q(\mathbf{r}) = q(\mathbf{s})$.

⟨2⟩1. $(\mathbf{r},\mathbf{s}) \neq 0$.
PROOF: The right side of ⟨1⟩ is positive by ⟨1⟩1; its left side cannot vanish. □

⟨2⟩2. $|u| = |v| = 1$.
PROOF: Apply ⟨1⟩2 with ⟨2⟩1. Its product identity and ⟨1⟩ give positive integers $|u|, |v|$ with product 1. Each is at least 1, so neither can exceed 1. □

⟨2⟩3. Q.E.D.
PROOF: Apply the ratio identity in ⟨1⟩2 with ⟨2⟩1. Substitute ⟨2⟩2 and multiply by $q(\mathbf{r}) > 0$ from ⟨1⟩1. □

⟨1⟩4. ASSUME: $2(\mathbf{r},\mathbf{s})^2 = q(\mathbf{r})q(\mathbf{s})$.
PROVE: $q(\mathbf{s}) = 2q(\mathbf{r})$ or $q(\mathbf{r}) = 2q(\mathbf{s})$.

⟨2⟩1. $(\mathbf{r},\mathbf{s}) \neq 0$.
PROOF: The right side of ⟨1⟩ is positive by ⟨1⟩1; its left side cannot vanish. □

LET: $A = |u|$ and $B = |v|$.

⟨2⟩2. A, B are positive integers, $AB = 2$ and $1 \leq A \leq 2$.
PROOF: Apply ⟨1⟩2 with ⟨2⟩1. Its product identity and ⟨1⟩ give $AB = 2$ and positivity; $B \geq 1$ bounds A by 2. □

⟨2⟩3. CASE: $A = 1$.
PROVE: $q(\mathbf{r}) = 2q(\mathbf{s})$.
PROOF: Step ⟨2⟩2 gives $B = 2$. Apply the ratio identity in ⟨1⟩2 with ⟨2⟩1, then clear the positive denominator from ⟨1⟩1. □

⟨2⟩4. CASE: $A = 2$.
PROVE: $q(\mathbf{s}) = 2q(\mathbf{r})$.
PROOF: Step ⟨2⟩2 gives $B = 1$. Apply the ratio identity in ⟨1⟩2 with ⟨2⟩1, then clear the positive denominator from ⟨1⟩1. □

⟨2⟩5. Q.E.D.
PROOF: The integer interval in ⟨2⟩2 makes ⟨2⟩3 and ⟨2⟩4 exhaustive. □

⟨1⟩5. Q.E.D.
PROOF: Steps ⟨1⟩3 and ⟨1⟩4 give the two conclusions. □

Lemma (integer logarithm shifts). ASSUME: $x \in \mathbb{R}$ and $x > 0$.
LET: $\eta(y) = \text{Int.log}(2, y)$ for real $y > 0$.
PROVE: $\eta(2x) = \eta(x) + 1$ and $\eta(x/2) = \eta(x) - 1$.

PROOF SKETCH: We will multiply the defining interval by 2, then apply the doubling identity at $x/2$.

⟨1⟩1. For every real $y > 0$, $\eta(2y) = \eta(y) + 1$.
⟨2⟩1. $2^{\eta(y)} \leq y < 2^{\eta(y)+1}$.

PROOF: `Int.zpow_log_le_self` and `Int.lt_zpow_succ_log_self` with base $2 > 1$ and $y > 0$ in ⟨1⟩1. □

⟨2⟩2. $2^{\eta(y)+1} \le 2y < 2^{(\eta(y)+1)+1}$.
PROOF: Multiply ⟨2⟩1 by $2 > 0$ and apply $\texttt{zpow_add_one}_0$ at base $2 \neq 0$. □

⟨2⟩3. $\eta(y) + 1 \le \eta(2y)$.
PROOF: Apply `Int.zpow_le_iff_le_log` at $2y > 0$ to the lower bound of ⟨2⟩2. □

⟨2⟩4. $\eta(2y) < \eta(y) + 2$.
PROOF: Apply `Int.lt_zpow_iff_log_lt` at $2y > 0$ to the upper bound of ⟨2⟩2. □

⟨2⟩5. Q.E.D.
PROOF: The adjacent integer bounds in ⟨2⟩3 and ⟨2⟩4. □

⟨1⟩2. $\eta(x/2) = \eta(x) - 1$.
PROOF: Apply ⟨1⟩1 to $x/2 > 0$, using ⟨0⟩: $\eta(x) = \eta(2(x/2)) = \eta(x/2) + 1$. Rearrange this integer equality. □

⟨1⟩3. Q.E.D.
PROOF: Step ⟨1⟩1 at $x > 0$ and ⟨1⟩2. □

The two logarithm lemmas are declarations in namespace `Int`; they concern positive real numbers, independently of line systems.

Lemma (colouring selected root lines). ASSUME: 1. V is a finite-dimensional real inner product space, IsRootSet(Φ), and $S \subseteq \mathcal{L}(\Phi)$.
2. HasAllowedAngles(S).
PROVE: HasBipartition(S).

PROOF SKETCH: We will colour a line by the parity of the integer logarithm of its squared root length. Orthogonal pairs impose no constraint.

⟨1⟩1. PICK $\mathbf{r}_\ell \in \Phi$ with $\ell = \mathbb{R}\mathbf{r}_\ell$ for every $\ell \in S$.
PROOF: Containment in ⟨0⟩:1 and Definition (lines of a vector set) give the witnesses. □

LET: $\delta(\ell) = q(\mathbf{r}_\ell)$, $\eta(x) = \mathrm{Int.log}(2, x)$, and $b(\ell) =$ true exactly when $\eta(\delta(\ell))$ is odd.

⟨1⟩2. Every $\ell \in S$ has dimension one, and $\delta(\ell) > 0$.
⟨2⟩1. $\mathbf{r}_\ell \neq 0$ and $\delta(\ell) > 0$.
PROOF: Definition 3.9, ⟨1⟩1, and positive definiteness. □
⟨2⟩2. $\dim \ell = 1$.
PROOF: Apply `finrank_span_singleton` to ⟨2⟩1 and ⟨1⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Steps ⟨2⟩1 and ⟨2⟩2. □

⟨1⟩3. ASSUME: $\ell, m \in S$, $\ell \neq m$, and HasSquaredCosine($\ell, m, 1/4$).
PROVE: $b(\ell) = b(m)$.
⟨2⟩1. $\delta(\ell) = \delta(m)$.
PROOF: Lemma (squared cosine from representatives), applied to ⟨1⟩ and the nonzero representatives from ⟨1⟩1 and ⟨1⟩2, gives $4(\mathbf{r}_\ell, \mathbf{r}_m)^2 = \delta(\ell)\delta(m)$. Apply part 1 of Lemma (lengths on root lines) to these roots and the Root System in ⟨0⟩:1. □
⟨2⟩2. Q.E.D.
PROOF: Equal inputs in ⟨2⟩1 give equal integer logarithms and equal Boolean colours. □

⟨1⟩4. ASSUME: $\ell, m \in S$, $\ell \neq m$, and HasSquaredCosine($\ell, m, 1/2$).
PROVE: $b(\ell) \neq b(m)$.
⟨2⟩1. $\delta(m) = 2\delta(\ell)$ or $\delta(m) = \delta(\ell)/2$.
PROOF: Lemma (squared cosine from representatives), applied to ⟨1⟩ and the representatives in ⟨1⟩1 and ⟨1⟩2, gives $2(\mathbf{r}_\ell, \mathbf{r}_m)^2 = \delta(\ell)\delta(m)$. Part 2 of Lemma (lengths on root lines), using ⟨0⟩:1, gives $\delta(m) = 2\delta(\ell)$ or $\delta(\ell) = 2\delta(m)$. Divide the second equality by 2. □

⟨2⟩2. CASE: $\delta(m) = 2\delta(\ell)$.
PROVE: $b(\ell) \neq b(m)$.
⟨3⟩1. $\eta(\delta(m)) = \eta(\delta(\ell)) + 1$.
PROOF: The doubling identity in Lemma (integer logarithm shifts) and positivity in ⟨1⟩2. □
⟨3⟩2. Q.E.D.
PROOF: Consecutive integers have opposite parities by Integer arithmetic and logarithms; apply ⟨3⟩1 and the definition of b. □
⟨2⟩3. CASE: $\delta(m) = \delta(\ell)/2$.
PROVE: $b(\ell) \neq b(m)$.
⟨3⟩1. $\eta(\delta(m)) = \eta(\delta(\ell)) - 1$.
PROOF: The halving identity in Lemma (integer logarithm shifts) and positivity in ⟨1⟩2. □
⟨3⟩2. Q.E.D.
PROOF: Integers differing by 1 have opposite parities by Integer arithmetic and logarithms; apply ⟨3⟩1 and the definition of b. □
⟨2⟩4. Q.E.D.
PROOF: Step ⟨2⟩1 makes ⟨2⟩2 and ⟨2⟩3 exhaustive. □
⟨1⟩5. ASSUME: $\ell, m \in S$, $\ell \neq m$, and $b(\ell) = b(m)$.
PROVE: HasSquaredCosine$(\ell, m, 0)$ or HasSquaredCosine$(\ell, m, 1/4)$.
PROOF: Hypothesis ⟨0⟩:2 and Definition (angles and compatible colourings) give a parameter in $\{0, 1/4, 1/2\}$ satisfying the predicate. Step ⟨1⟩4 and the colour equality in ⟨1⟩ exclude the predicate at $1/2$. The other two parameters give the stated disjunction. □
⟨1⟩6. ASSUME: $\ell, m \in S$, $\ell \neq m$, and $b(\ell) \neq b(m)$.
PROVE: HasSquaredCosine$(\ell, m, 0)$ or HasSquaredCosine$(\ell, m, 1/2)$.
PROOF: Hypothesis ⟨0⟩:2 and Definition (angles and compatible colourings) give a parameter in $\{0, 1/4, 1/2\}$ satisfying the predicate. Step ⟨1⟩3 and the colour inequality in ⟨1⟩ exclude the predicate at $1/4$. The other two parameters give the stated disjunction. □
⟨1⟩7. Q.E.D.
PROOF: Steps ⟨1⟩5 and ⟨1⟩6 give the two clauses of Definition (angles and compatible colourings); b is the required witness. For $S = \varnothing$ it is the empty function. □

Theorem 3.12. ASSUME: 1. V is a finite-dimensional real inner product space. The set S consists of real subspaces of V, with $\dim \ell = 1$ for every $\ell \in S$.
2. HasAllowedAngles(S).
PROVE:

$$\big(\exists \Phi \subseteq V, \quad \text{IsRootSet}(\Phi) \text{ and } S \subseteq \mathcal{L}(\Phi)\big) \quad \Longleftrightarrow \quad \text{HasBipartition}(S).$$

PROOF SKETCH: **[repair]** The paper gives no proof (Gap 11). We will use the normalized construction for sufficiency and the integer-logarithm colouring for necessity.
⟨1⟩1. ASSUME: There is a Root System Φ in V with $S \subseteq \mathcal{L}(\Phi)$.
PROVE: HasBipartition(S).
PROOF: Lemma (colouring selected root lines) with the finite dimension in ⟨0⟩:1, the containment and Root System in ⟨1⟩, and the angle hypothesis ⟨0⟩:2. □
⟨1⟩2. ASSUME: HasBipartition(S).
PROVE: There is a Root System Φ in V with $S \subseteq \mathcal{L}(\Phi)$.
PROOF: Part 1 of Corollary (normalized containment), with ⟨0⟩:1 and ⟨1⟩. □
⟨1⟩3. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2 are the two implications. □

Conventions in the statement. **[repair]** A bipartition means the two fibres of a Boolean colouring; either fibre may be empty. This is required even for a single line, or for two lines at 60°, which must

have the same colour. The ordinary proof records this as Implicit hypothesis 8 of `cgss/errata.md`. The word *and* between the two cross-part angles in the printed statement is read as *or* (Error 13, item 3). Finiteness is a conclusion of Lemma (finitely many lines); spanning and nonemptiness of S are not hypotheses. Corollary (normalized containment) separately states the stronger normalized construction, in both the span and the given ambient space. Allowing a containing Root System in a larger finite-dimensional inner product space gives the same condition: apply Lemma (colouring selected root lines) in that space to the isometric images of the selected lines and pull back its colouring.

The Lean declaration has argument `S:Set(SubmoduleRealV)`, not an argument of the bundled type `LineSystemV`, which permits only the 60° and 90° cases. Its ambient assumptions are `NormedAddCommGroupV`, `InnerProductSpaceRealV`, and `FiniteDimensionalRealV`; the line hypothesis is `Module.finrankReal` equal to 1 on each member of S. The docstring records the empty-fibre convention and the errata references above. The names in the metadata are the settled ones from `cgss/notation.md`, Section 10, supplemented by the separately stated helper lemmas.

1.5. **The bridge from Section 2.**

Lemma (the lines of a line system). ASSUME: V is a finite-dimensional real inner product space and $R : \mathrm{LineSystem}(V)$ is as in Definition (line systems of Section 2).
PROVE: $\mathrm{HasAllowedAngles}(\mathcal{L}(R.\Phi))$ and $\mathrm{HasBipartition}(\mathcal{L}(R.\Phi))$, witnessed by the constant colouring with value false.

PROOF SKETCH: We will exclude the two endpoint inner products on distinct lines and then use the constant colouring.

⟨1⟩1. Every $\ell \in \mathcal{L}(R.\Phi)$ has dimension one.
PROOF: A representative has squared norm 2 by Definition (line systems of Section 2), so is nonzero; apply `finrank_span_singleton` and Definition (lines of a vector set). □

⟨1⟩2. ASSUME: $\ell, m \in \mathcal{L}(R.\Phi)$ and $\ell \neq m$.
PROVE: $\mathrm{HasSquaredCosine}(\ell, m, 0)$ or $\mathrm{HasSquaredCosine}(\ell, m, 1/4)$.

⟨2⟩1. PICK $\mathbf{r}, \mathbf{s} \in R.\Phi$ with $\ell = \mathbb{R}\mathbf{r}$, $m = \mathbb{R}\mathbf{s}$.
PROOF: Definition (lines of a vector set). □

⟨2⟩2. ASSUME: $(\mathbf{r}, \mathbf{s}) = 2$.
PROVE: False.

⟨3⟩1. $q(\mathbf{r} - \mathbf{s}) = 0$ and $\mathbf{r} = \mathbf{s}$.
PROOF: Bilinearity and the squared norms in Definition (line systems of Section 2) give $q(\mathbf{r} - \mathbf{s}) = 2 + 2 - 2 \cdot 2 = 0$; positive definiteness gives the equality. □

⟨3⟩2. Q.E.D.
PROOF: Step ⟨3⟩1 makes the spans in ⟨2⟩1 equal, contrary to ⟨1⟩. □

⟨2⟩3. ASSUME: $(\mathbf{r}, \mathbf{s}) = -2$.
PROVE: False.

⟨3⟩1. $q(\mathbf{r} + \mathbf{s}) = 0$ and $\mathbf{r} = -\mathbf{s}$.
PROOF: Bilinearity gives $q(\mathbf{r} + \mathbf{s}) = 2 + 2 + 2(-2) = 0$; positive definiteness gives the equality. □

⟨3⟩2. Q.E.D.
PROOF: Multiplication by -1 does not change a real span; ⟨3⟩1 contradicts the distinct spans in ⟨1⟩. □

⟨2⟩4. $(\mathbf{r}, \mathbf{s}) \in \{-1, 0, 1\}$.
PROOF: Exclude the two endpoint values in Definition (line systems of Section 2) by ⟨2⟩2 and ⟨2⟩3. □

⟨2⟩5. Q.E.D.
PROOF: Step ⟨2⟩4 gives $(\mathbf{r},\mathbf{s})^2 \in \{0,1\}$. The squared norms are 2 by Definition (line systems of Section 2). Lemma (squared cosine from representatives), using ⟨1⟩1 and ⟨2⟩1, gives the predicate at 0 or 1/4. □

⟨1⟩3. HasAllowedAngles($\mathcal{L}(R.\Phi)$).
PROOF: Step ⟨1⟩2 supplies an allowed parameter in Definition (angles and compatible colourings). □

⟨1⟩4. The constant function $b(\ell) = \text{false}$ is compatible.
⟨2⟩1. For distinct ℓ, m, the equal-colour angle clause holds.
PROOF: Step ⟨1⟩2. □
⟨2⟩2. There are no ℓ, m with $b(\ell) \neq b(m)$.
PROOF: The definition of the constant function. □
⟨2⟩3. Q.E.D.
PROOF: Steps ⟨2⟩1 and ⟨2⟩2 give the two clauses of Definition (angles and compatible colourings). □

⟨1⟩5. Q.E.D.
PROOF: Steps ⟨1⟩3 and ⟨1⟩4. □

The exact computational examples remain in the ordinary proof and its `cgss/proofs/thm312/checks/check_thm312.py`. They are not premises of any result here. The construction and colouring use neither the classification of Root Systems nor a two-length assertion for a reducible Root System. The proof permits unrelated scales on orthogonal components, and restricts angles only on the selected set of lines.

REFERENCES

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